NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE | NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

3.1 Probability and Statistics for Machine Learning

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

Lesson Objectives

Understand and apply the concepts of probability, information theory and statistics

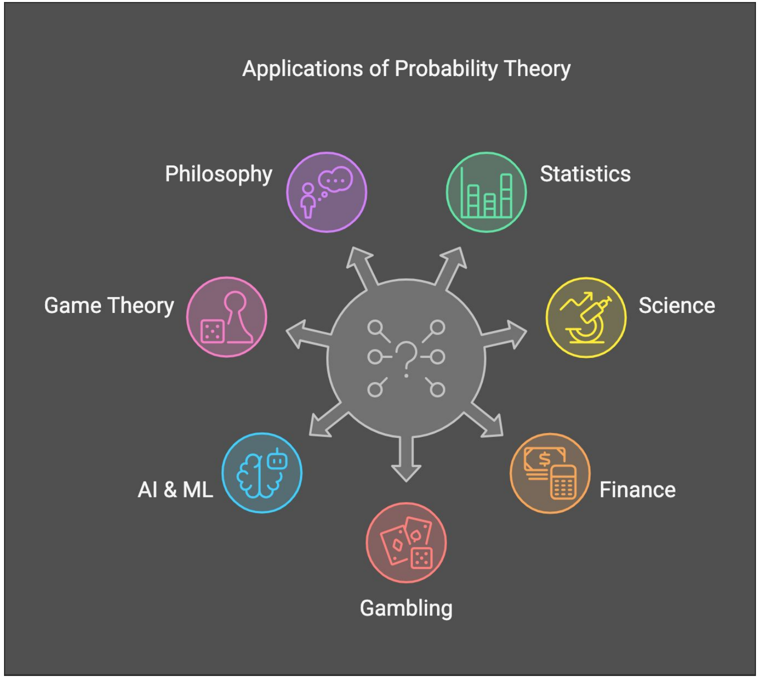
Lesson Plan

1. Introduction to probability
2. Distributions in Machine Learning
3. Introduction to Statistics

What is probability theory ?

Probability theory is the mathematical framework for quantifying uncertainty. It helps us answer questions like "What are the chances?" and provides the foundation for making informed decisions when outcomes are uncertain.

- ☔ **Weather Prediction**: Predicting the likelihood of rain.
- 🎲 **Gambling**: Calculating odds in poker or roulette
- 📉 **Finance**: Assessing potential risks and returns on investments
- 🩻 **Medical Diagnosis**: Evaluating the probability of a disease given certain symptoms



Definitions

Event

A subset of outcomes that we are interested in measuring

Sample Space (Ω)

The set of all possible outcomes in an experiment

Example: Rolling a die

Sample Space (Ω): *{1, 2, 3, 4, 5, 6}*

Event (E): Getting an even number *{2, 4, 6}*

Law of Large Numbers

"As the number of trials increases, the observed probability converges to the true probability"

Example:

As the number of coin flips increases, the observed probability of heads will converge to 50%.

- Foundation for probability and statistics.
- Relationship between **theoretical** probability and **empirical** results.

Jupyter Notebook Exercises

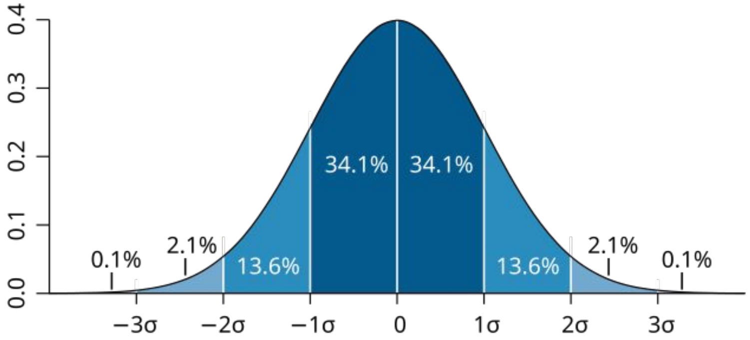
Part 1

- **Simulating Coin Tosses**: Use Python to demonstrate the law of large numbers.
- **Measures of Central Tendency**: Mean, Median and Mode

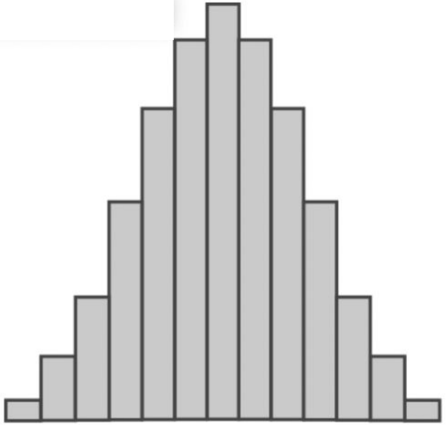
Probability Distributions

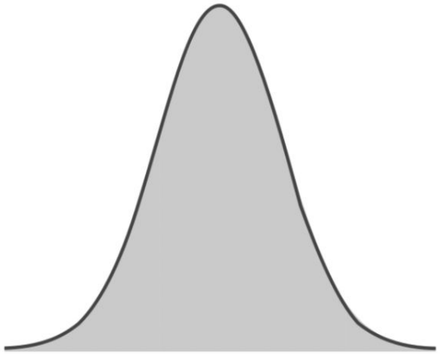
🎲 Describes how likely different outcomes of a random event to occur.

🗺️ A map that shows all possible results and how often each should occur.



There are two main types:
Discrete and **Continuous** PDs





Probability Distributions: The Shape of Uncertainty

Probability distributions show us how likely different outcomes are. They're the building blocks for understanding patterns in data and making predictions about future events.

Discrete Distributions

Deal with countable outcomes like rolling dice or counting defective products. Examples include binomial and Poisson distributions.

- Binomial: success/failure scenarios
- Poisson: rare events over time

Continuous Distributions

Handle measurements that can take any value within a range, like height or temperature. The normal distribution is the most famous example.

- Normal: bell-shaped, symmetric
- Uniform: equally likely outcomes

Discrete Probability Distribution

Apply to **countable** outcomes (like whole numbers).

Each outcome has a specific probability and all probabilities add up to 1.

E.g. *Binomial, Poisson, Geometric* distributions

Binomial Distribution
(success/failure trials)

Flipping a coin 3 times (n = 3 trials)

For example, P(2 heads) = 3/8

Binomial Distribution

The **Binomial Probability Formula** is:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{(n-k)}$$

where:

- n = total trials
- k = number of successes
- p = probability of success per trial
- $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ (combinations)

The **binomial coefficient**, often written as **"n choose k"** and denoted as:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

is used to determine the **number of ways to choose** k items from n distinct items **without considering order**.

Binomial Distribution

For $n = 3$, $k = 2$, and $p = 0.5$:

$$P(X = 2) = \binom{3}{2}(0.5)^2(1 - 0.5)^{3-2}$$

$$= \frac{3!}{2!(3-2)!}(0.5)^2(0.5)^1$$

$$= \frac{3!}{2!1!}(0.5)^2(0.5)^1$$

$$= \frac{3 \times 2 \times 1}{2 \times 1 \times 1} \times 0.25 \times 0.5$$

$$= 3 \times 0.25 \times 0.5$$

$$= 3 \times 0.125$$

$$= 0.375$$

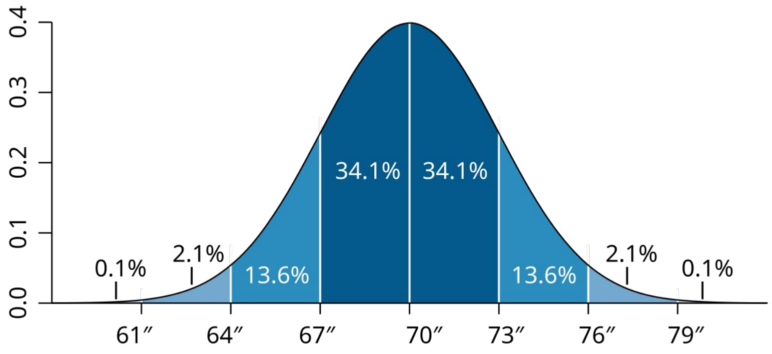
Thus, the probability of getting exactly **2 heads in 3 coin flips** is **0.375 or** $\frac{3}{8}$.

Flipping a coin 3 times (n = 3 trials)

For example, P(2 heads) = 3/8

Continuous Probability Distribution

- Apply to **uncountable** outcomes, e.g. measurements like time, height.
- Probabilities are calculated for ranges (e.g., 61”–79”), not single points.
- The total area under the curve equals 1.
- E.g. *Normal, Exponential* distributions



Normal Distribution

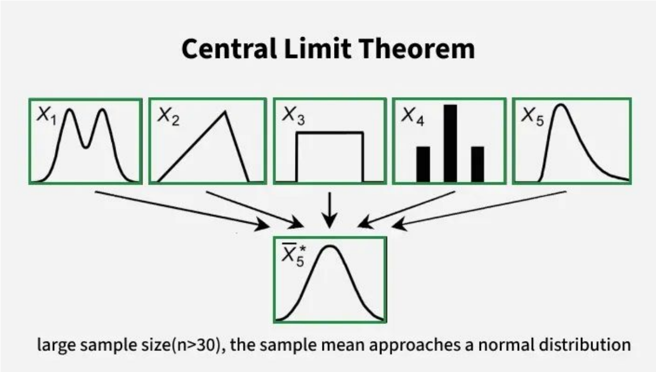
Most people cluster around the average (70” or 178cm), with fewer at extremes.

Mean = 70”, Standard Deviation = 3”.

Central Limit Theorem: The Magic of Large Numbers

The Central Limit Theorem is one of statistics' most powerful insights. It tells us that **when we take many samples from any population, the averages of those samples will form a normal distribution**.

This works regardless of the original population's shape. Whether you're sampling heights, test scores, or manufacturing defects, **sample means become predictably normal as sample size increases**.



> "The Central Limit Theorem is the reason we can make reliable inferences about populations from relatively small samples."

01

Take Multiple Samples

Collect several random samples from your population

02

Calculate Sample Means

Find the average of each individual sample

03

Observe the Pattern

Sample means form a normal distribution around the true population mean

Jupyter Notebook Exercises

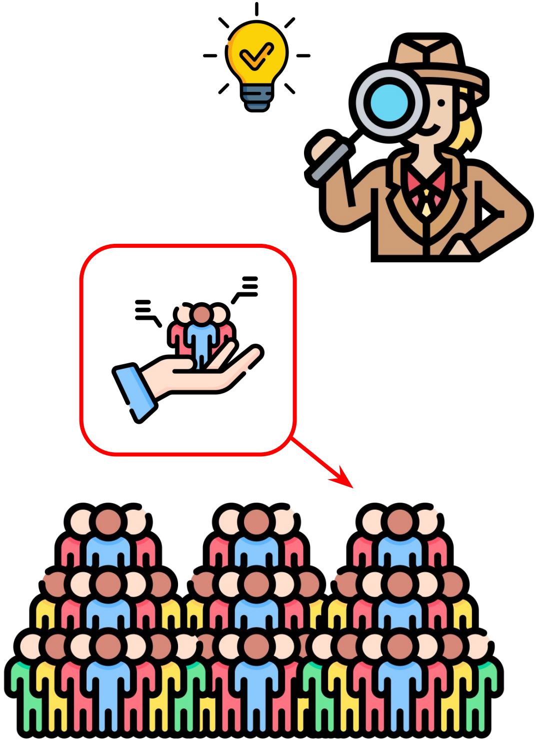
Part 2

- **Probability Distributions**: Generate and visualize distributions in Python.

Statistical Inference

Using probability to make conclusions about a population based on a sample.

- When studying a large population, it is impractical to collect data from every individual unit.
- Instead, study a smaller sample and use the results to draw conclusions about the whole population.



Statistical Inference

Statistical inference lets us make educated guesses about entire populations based on sample data. It's like being a detective, using clues from a small group to solve mysteries about a much larger group.

1

Sample Data

What we observe and measure from a subset of the population

2

Statistical Tests

Mathematical tools that help us analyze the evidence

3

Population Conclusions

Informed decisions about the larger group we're studying

Z-Scores

Z-scores transform any normal distribution into a standard normal distribution with mean 0 and standard deviation 1.

The Formula

$Z = (X - \mu) / \sigma$

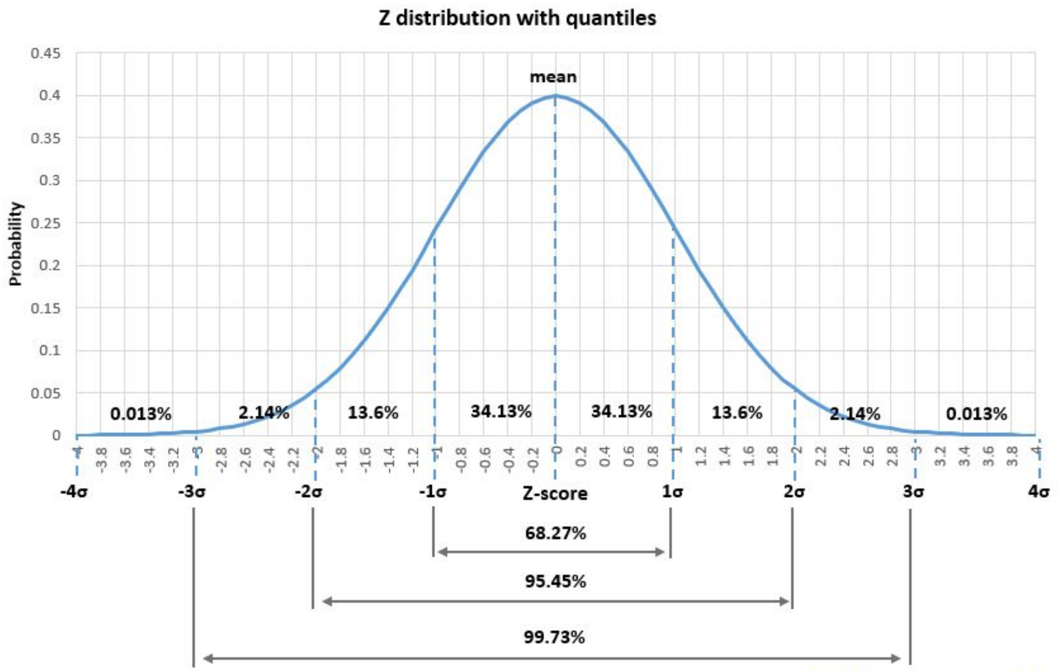
Where X is your value, μ is the mean, and σ is the standard deviation

Practical Use

Compare test scores across different exams, or determine how unusual a measurement is within its distribution

Interpretation

Z-scores tell you how many standard deviations away from the mean your value lies



https://www.gigacalculator.com/calculators/z-score-calculator.php

P-Value

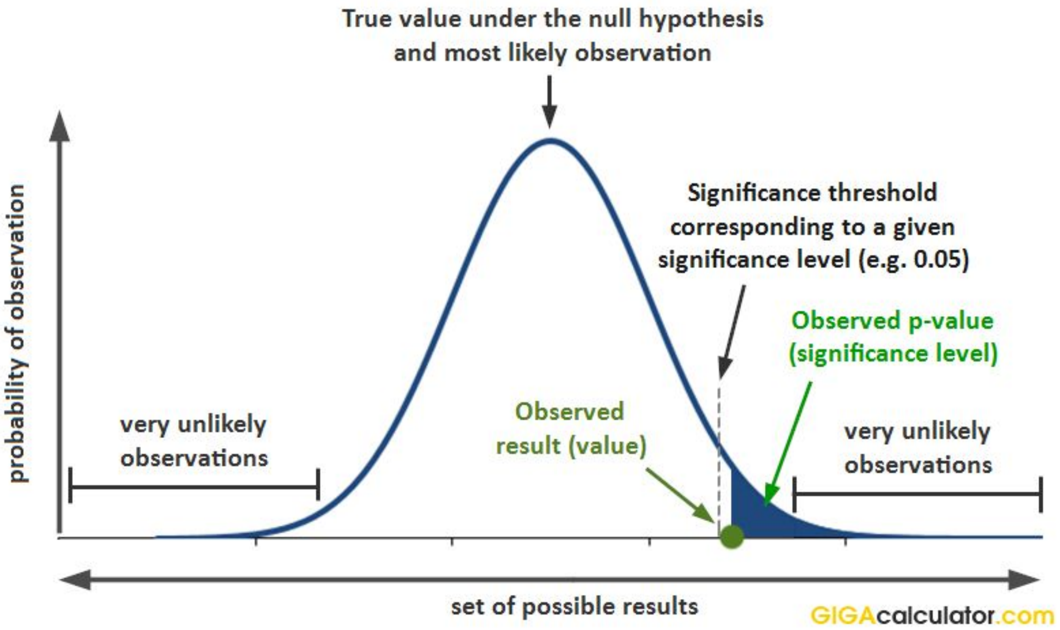
The p-value answers a crucial question: "If there really was no effect, what's the probability of seeing results this extreme or more extreme by pure chance?"

P < 0.05

Statistically significant - less than 5% chance results are due to random luck

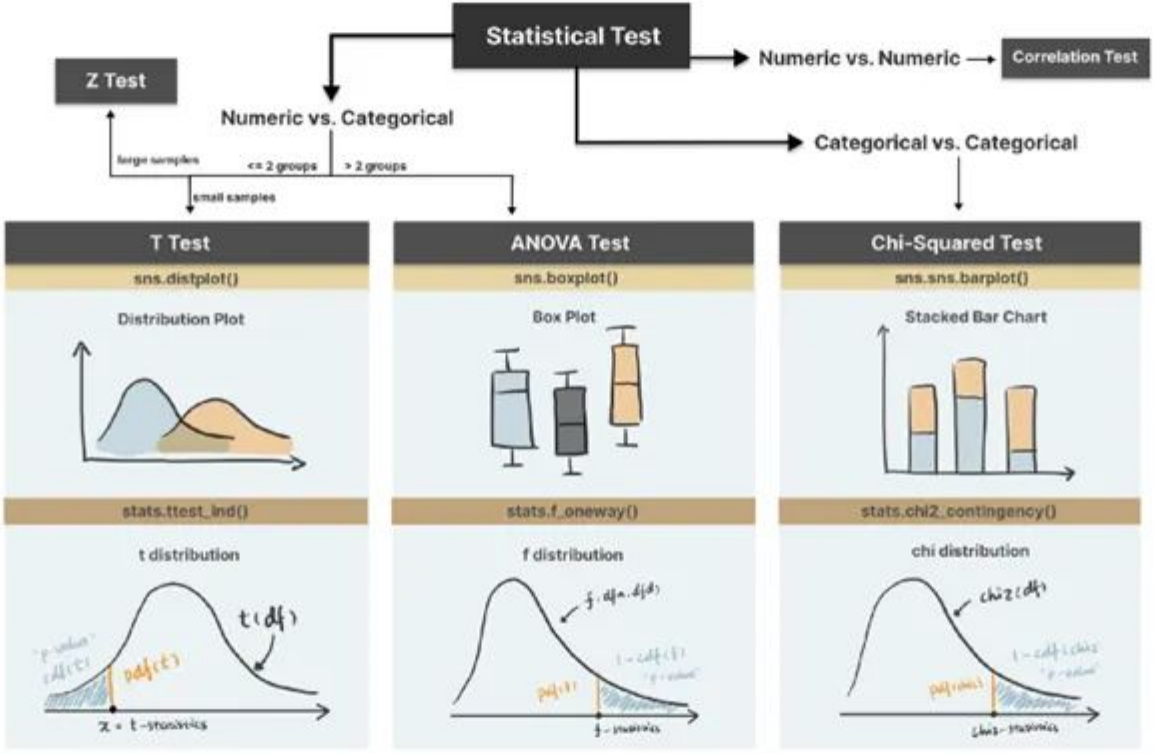
P > 0.05

Not statistically significant - could easily be due to chance



https://www.gigacalculator.com/articles/p-value-definition-and-interpretation-in-statistics/

Common Statistical Tests



https://medium.com/@hasninemirza/guidance-to-hypothesis-testing-in-python-t-test-anova-chi-squared-test-b2446ce44030

T-Test

Compares means between groups. Use when comparing average test scores between two classes or before/after treatment effects.

ANOVA

Compares means across multiple groups simultaneously. Perfect for comparing effectiveness of three or more different treatments.

Chi-Square Test

Tests relationships between categorical variables. Use to see if gender relates to product preference or if education level affects voting patterns.

Hypothesis Testing: The Scientific Method in Action

Hypothesis testing provides a systematic framework for making decisions under uncertainty. It's the foundation that supports all the statistical tests we've discussed.

1

State Your Hypotheses

Null hypothesis (H_0): No effect exists

Alternative hypothesis (H_1): An effect exists

2

Collect Data & Calculate Test Statistic

Use appropriate test (t-test, ANOVA, chi-square) based on your data type and research question

3

Choose Significance Level

Typically $\alpha = 0.05$, representing your tolerance for false positives

4

Make Your Decision

Compare p-value to α: reject H_0 if $p < \alpha$, fail to reject if $p \geq \alpha$

Key Takeaway: Statistics and probability give us powerful tools to make informed decisions in the face of uncertainty. From business analytics to medical research, these concepts help us separate signal from noise in our complex world.

Jupyter Notebook Exercises

Part 3

- **Hypothesis Test**: Walkthrough of statistical testing in Python.

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